

Lecture 01

Riemann Sums and the Definite Integral

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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
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Riemann Sums

The Definite Integral

Examples

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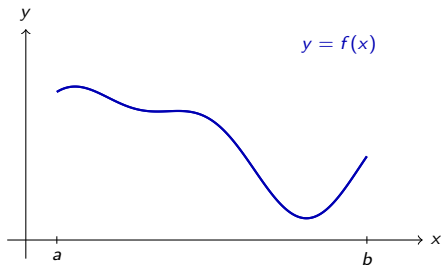
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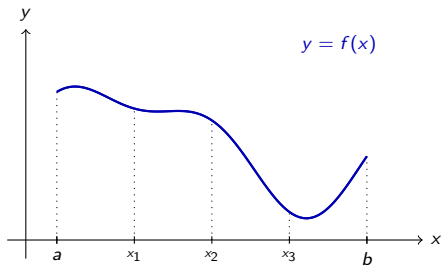
- We can choose the sub-interval lengths to be different too. The **norm** of a partition P , denoted $\|P\|$, is the length of the largest sub-interval.

From a Curve to the Riemann Sum: Step by Step



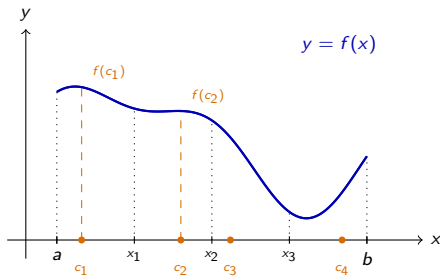
An arbitrary bounded function f on $[a, b]$.

From a Curve to the Riemann Sum: Step by Step



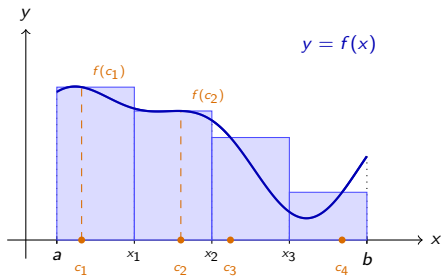
Split $[a, b]$ into subintervals: a partition $P = \{x_0, \dots, x_4\}$.

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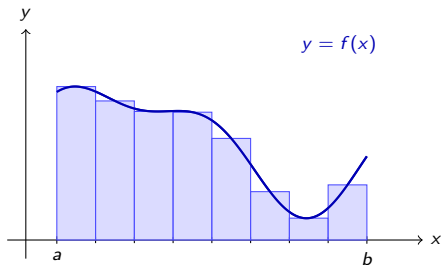
In each subinterval, choose a point c_k .

From a Curve to the Riemann Sum: Step by Step



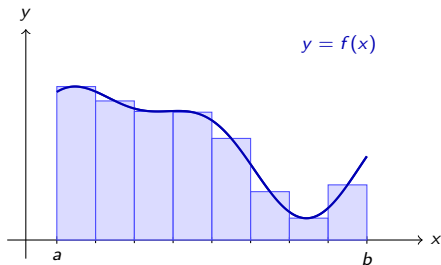
Build a rectangle of height $f(c_k)$ over each subinterval: $S_P = \sum_{k=1}^4 f(c_k) \Delta x_k$.

From a Curve to the Riemann Sum: Step by Step



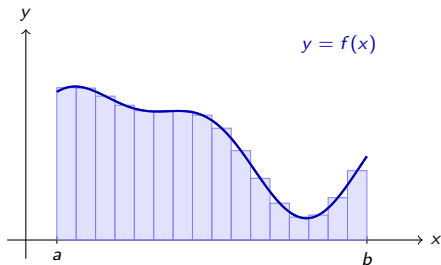
Refine the partition: more, thinner subintervals.

From a Curve to the Riemann Sum: Step by Step



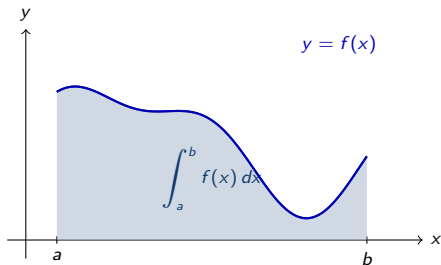
The Riemann sum with the finer partition approximates the area better.

From a Curve to the Riemann Sum: Step by Step



Keep refining $P \dots$

From a Curve to the Riemann Sum: Step by Step



As $\|P\| \rightarrow 0$, the rectangles' total area converges to $\int_a^b f(x) dx$.

From a Curve to the Riemann Sum: Step by Step

Definition

Consider $f : [a, b] \rightarrow \mathbb{R}$. We say f is Riemann integrable, and write $\int_a^b f(x) dx = L$, if for every $\epsilon > 0$ there exists $\delta > 0$ such that for every partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of $c_k \in [x_{k-1}, x_k]$,

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- If the sub-intervals have equal length,

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \frac{b-a}{n}.$$

Example: Integrable and Non-Integrable Functions

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- Consider the **Dirichlet function** $f : [0, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

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- **Why?** For any partition, choosing all c_k rational gives $S_P = b - a$; choosing all c_k irrational gives $S_P = 0$. The Riemann sum does not converge to a single value L .

- Partition $[a, b]$, choose sample points $c_k \in [x_{k-1}, x_k]$, and form the **Riemann sum**

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- Simple functions like $f(x) = 1$, $f(x) = x$, $f(x) = x^2$ are Riemann integrable, but not every bounded function is – the **Dirichlet function** is a standard counterexample.